```
Tanzania, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -34667.1            -34658.0                 0
     2 20263017             5000           0.276             -34660.4            -34648.6                 0
     3 20263018             5000           0.277             -34683.3            -34659.5                 0
     4 20263019             5000           0.276             -34670.7            -34652.4                 0
                                      file
 mcmc_tanzania_step2_chain1_2026-10-19.Rdata
 mcmc_tanzania_step2_chain2_2026-10-20.Rdata
 mcmc_tanzania_step2_chain3_2026-10-20.Rdata
 mcmc_tanzania_step2_chain4_2026-10-20.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
                                   trate.3                                    trate.1
                                     2.406                                      1.897
             hiv.specific.mortality.rates.0      age.40.to.49.transmission.multiplier.2
                                     1.856                                      1.709
             hiv.specific.mortality.rates.2          over.50.hiv.mortality.multiplier.0
                                     1.687                                      1.656
                                   trate.2     age.25.to.49.hiv.mortality.multiplier.0
                                     1.594                                      1.555
   age.50.and.over.transmission.multiplier.0              hiv.specific.mortality.rates.1
                                     1.509                                      1.504
      age.15.to.19.transmission.multiplier.1      age.40.to.49.transmission.multiplier.3
                                     1.494                                      1.387
            male.hiv.mortality.multiplier.1      age.30.to.39.transmission.multiplier.2
                                     1.370                                      1.365
      age.30.to.39.transmission.multiplier.3 age.65.to.79.mortality.slope.multiplier.male
                                     1.364                                      1.361
      age.5.to.14.hiv.mortality.multiplier.0          over.50.hiv.mortality.multiplier.1
                                     1.360                                      1.340
      age.20.to.29.transmission.multiplier.0  over.80.mortality.intercept.multiplier.male
                                     1.331                                      1.324
      age.20.to.29.transmission.multiplier.1                          age.assortativity
                                     1.320                                      1.310
      age.20.to.29.transmission.multiplier.3      age.40.to.49.transmission.multiplier.1
                                     1.300                                      1.284
   age.50.and.over.transmission.multiplier.3
                                     1.261

Merged simset:
 location simulations total.weight                                          file
 Tanzania        1000         1000 simset_tanzania_step2_chains1234_2026-10-20.Rdata
```

Tanzania step 2: likelihood

0
−10000
−20000
−30000

lue

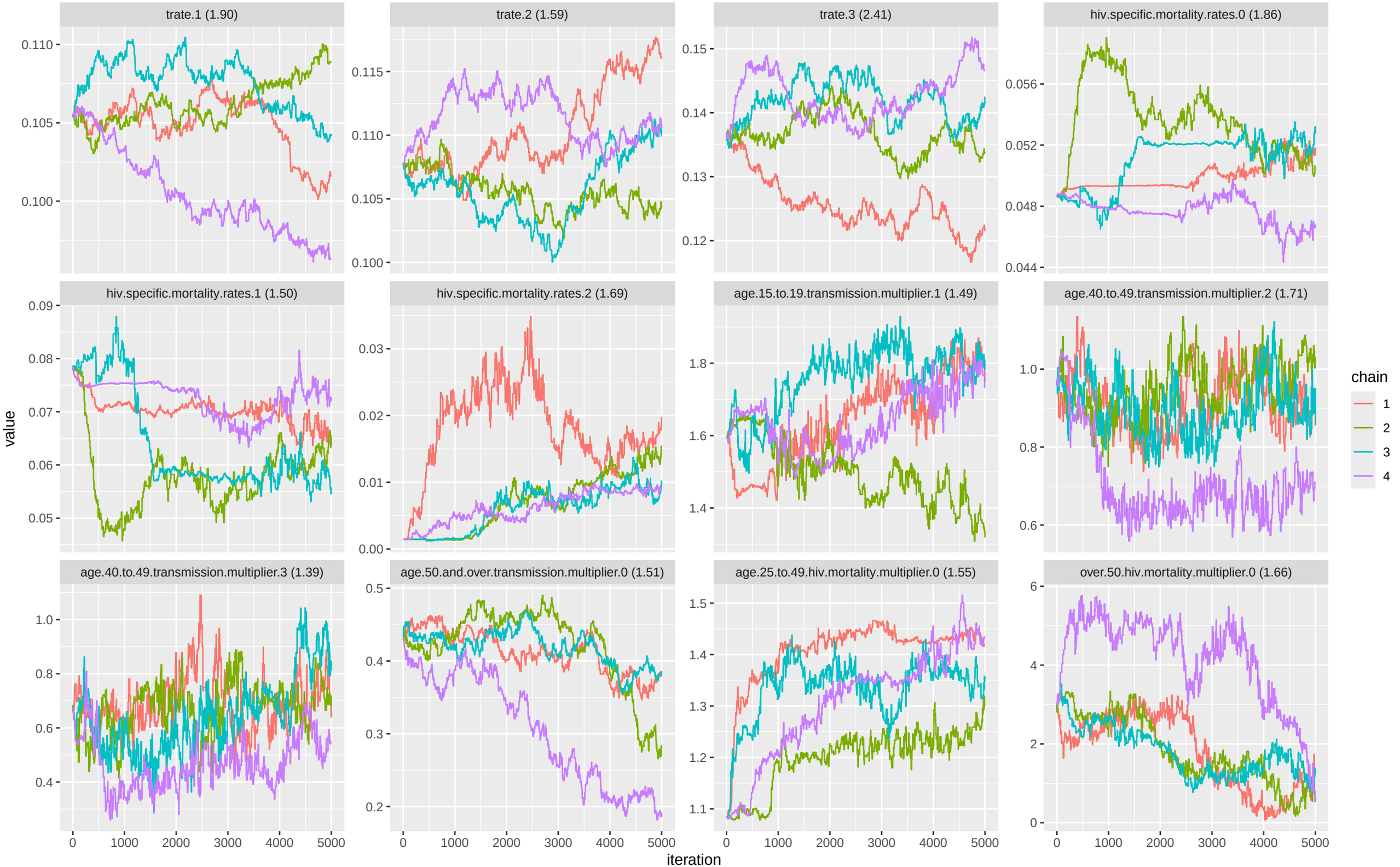
0
25000
50000
75000
100000

iteration

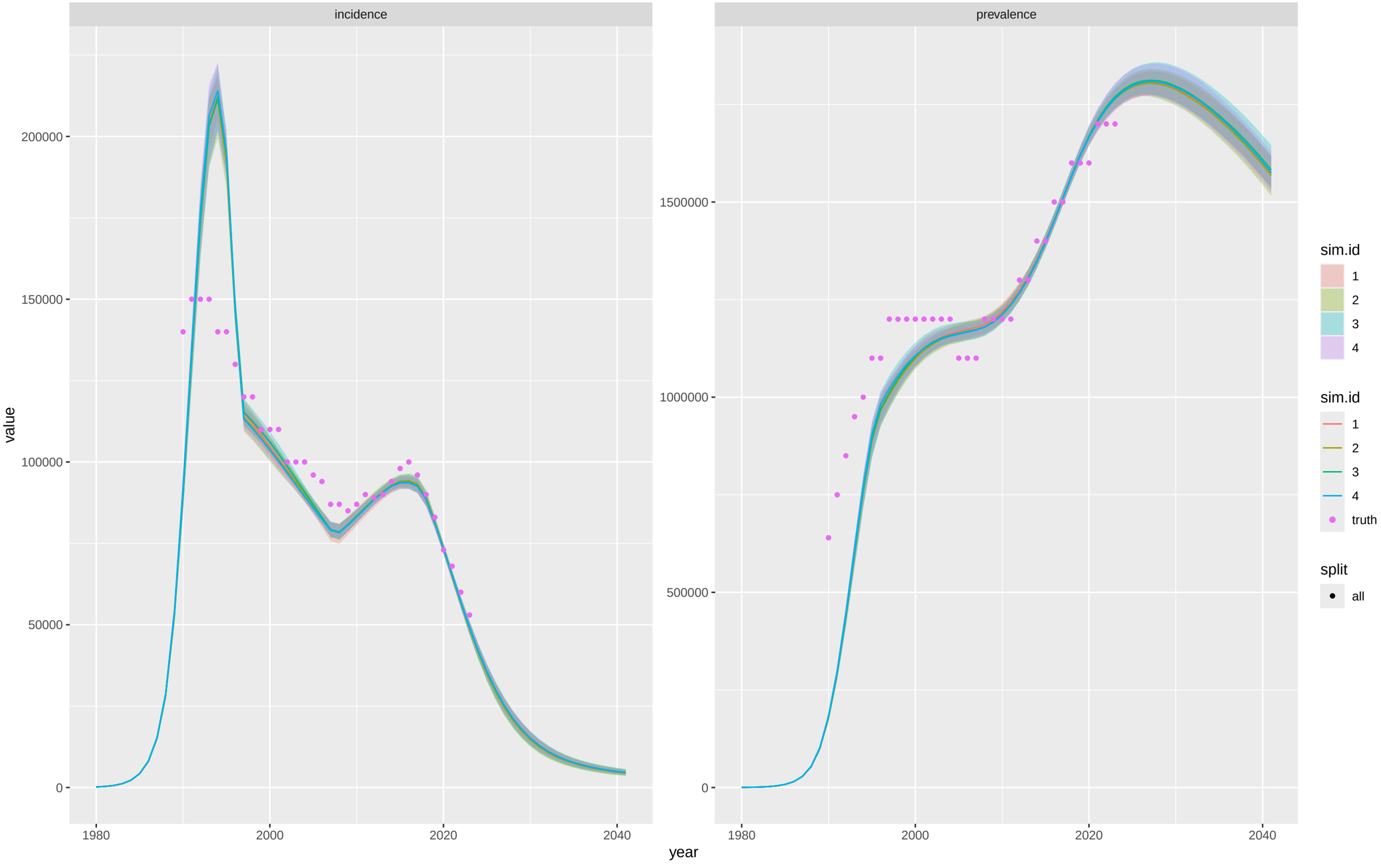
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

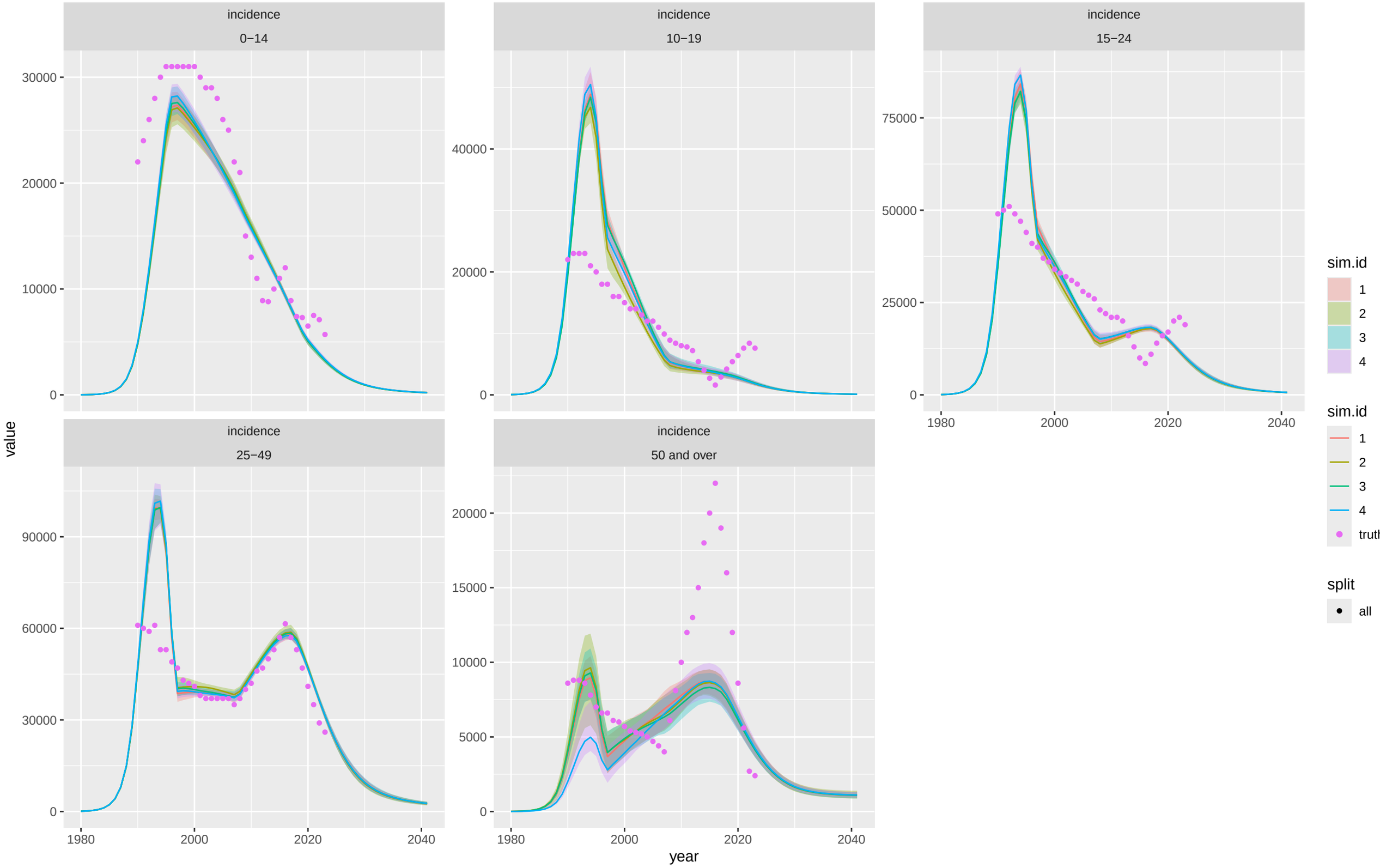
Tanzania step 2: trace



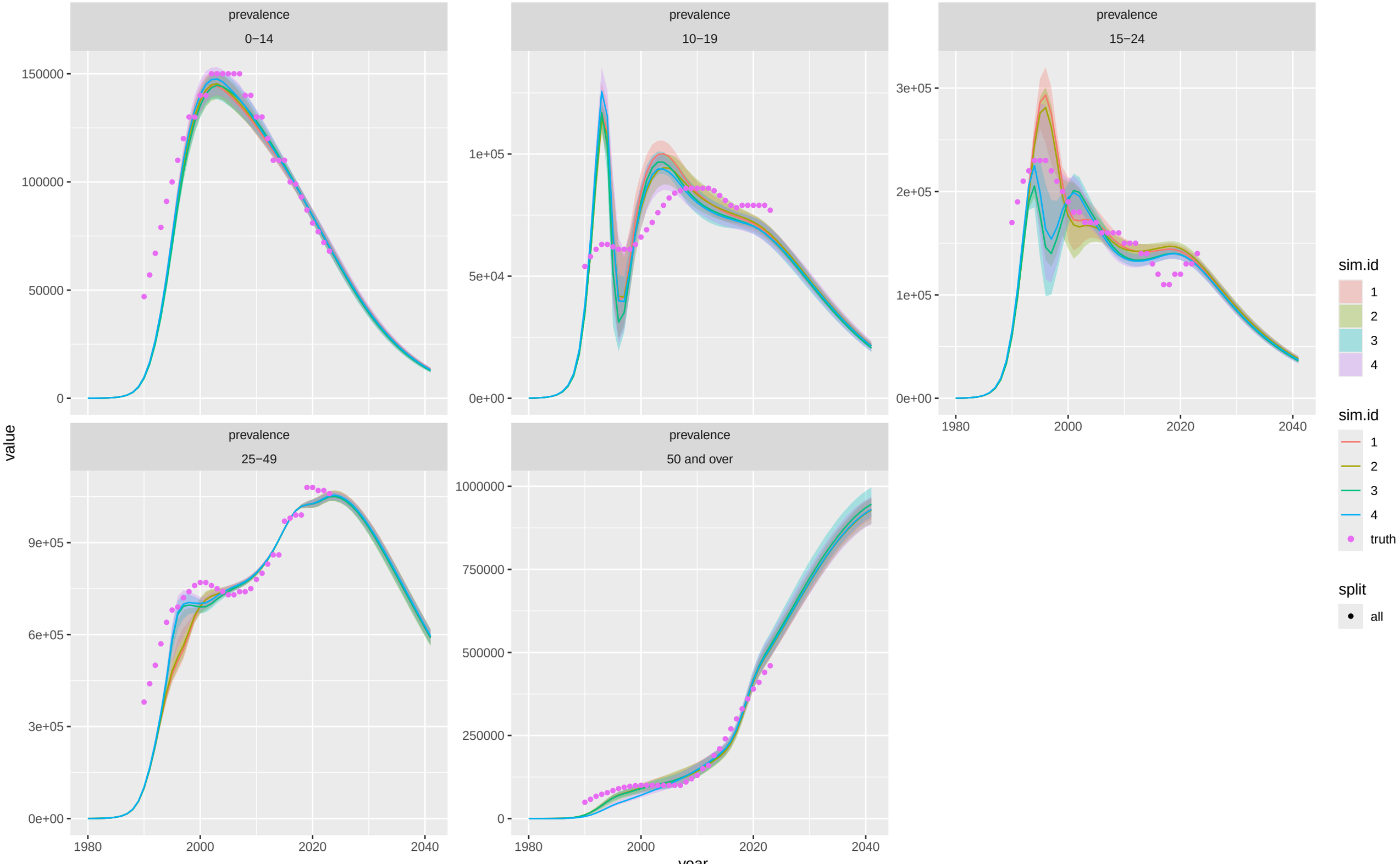
Tanzania step 2: fit (one color per chain)



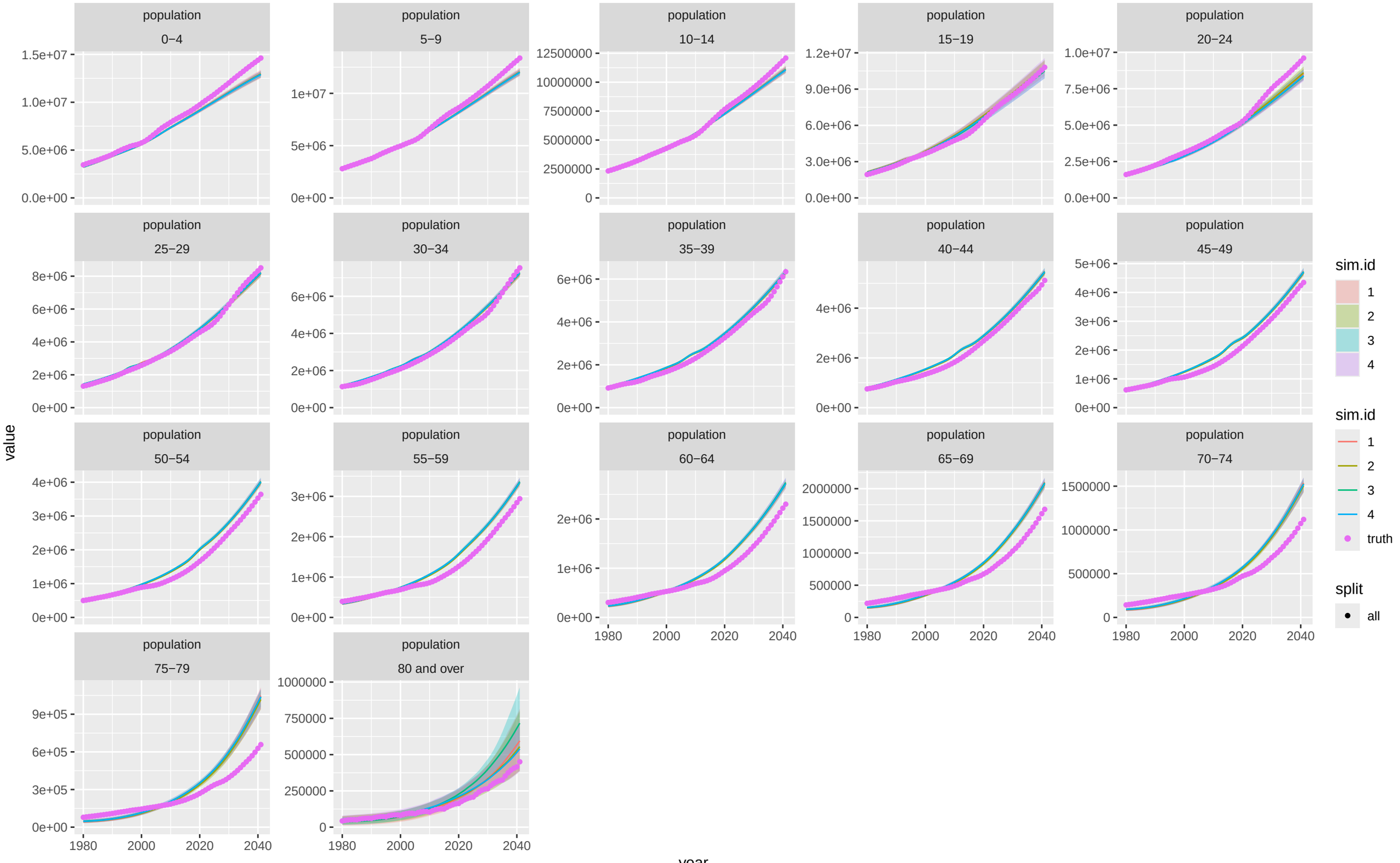
Tanzania step 2: incidence.by.age (one color per chain)



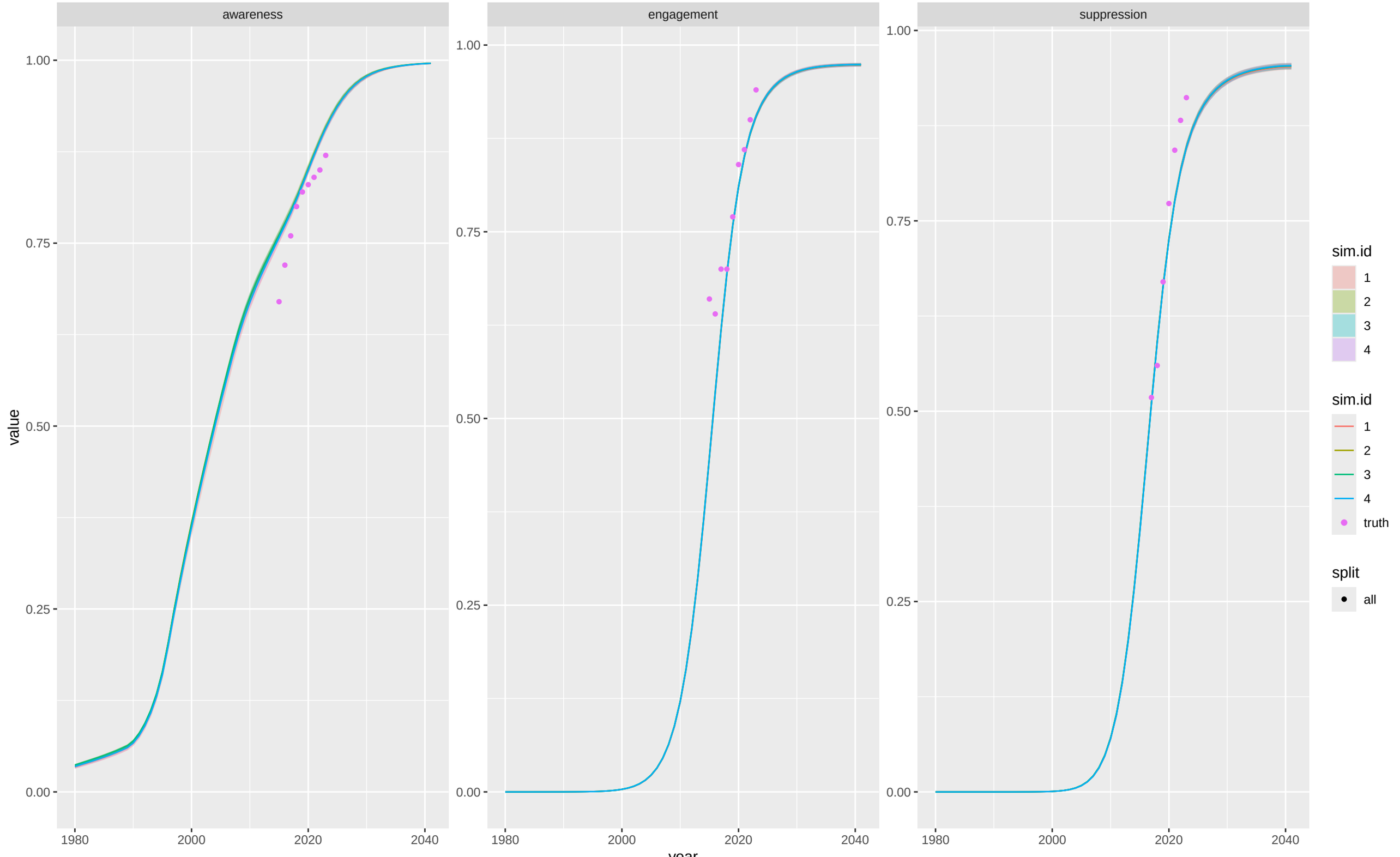
Tanzania step 2: prevalence.by.age (one color per chain)



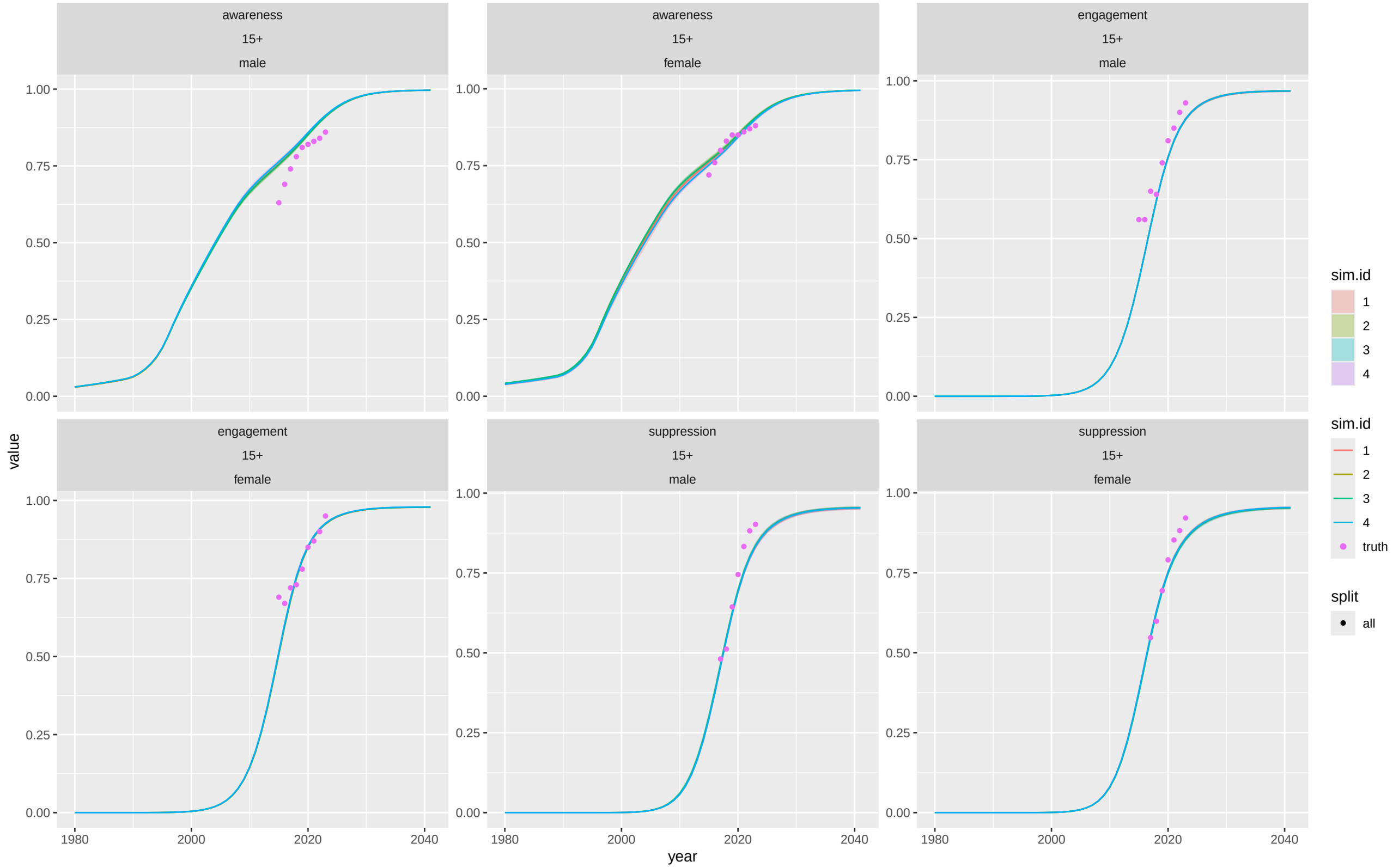
Tanzania step 2: population.by.age (one color per chain)



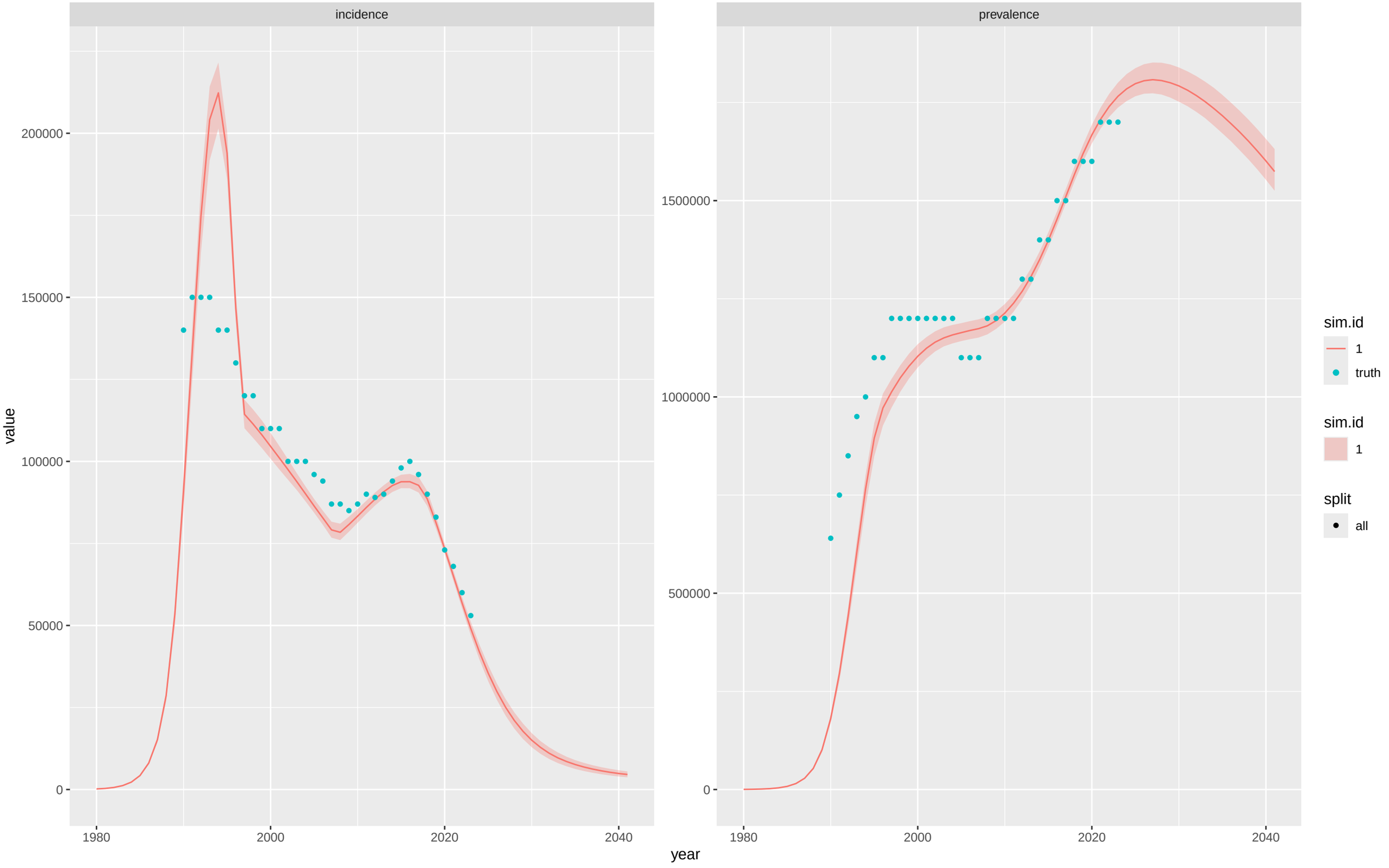
Tanzania step 2: cascade (one color per chain)



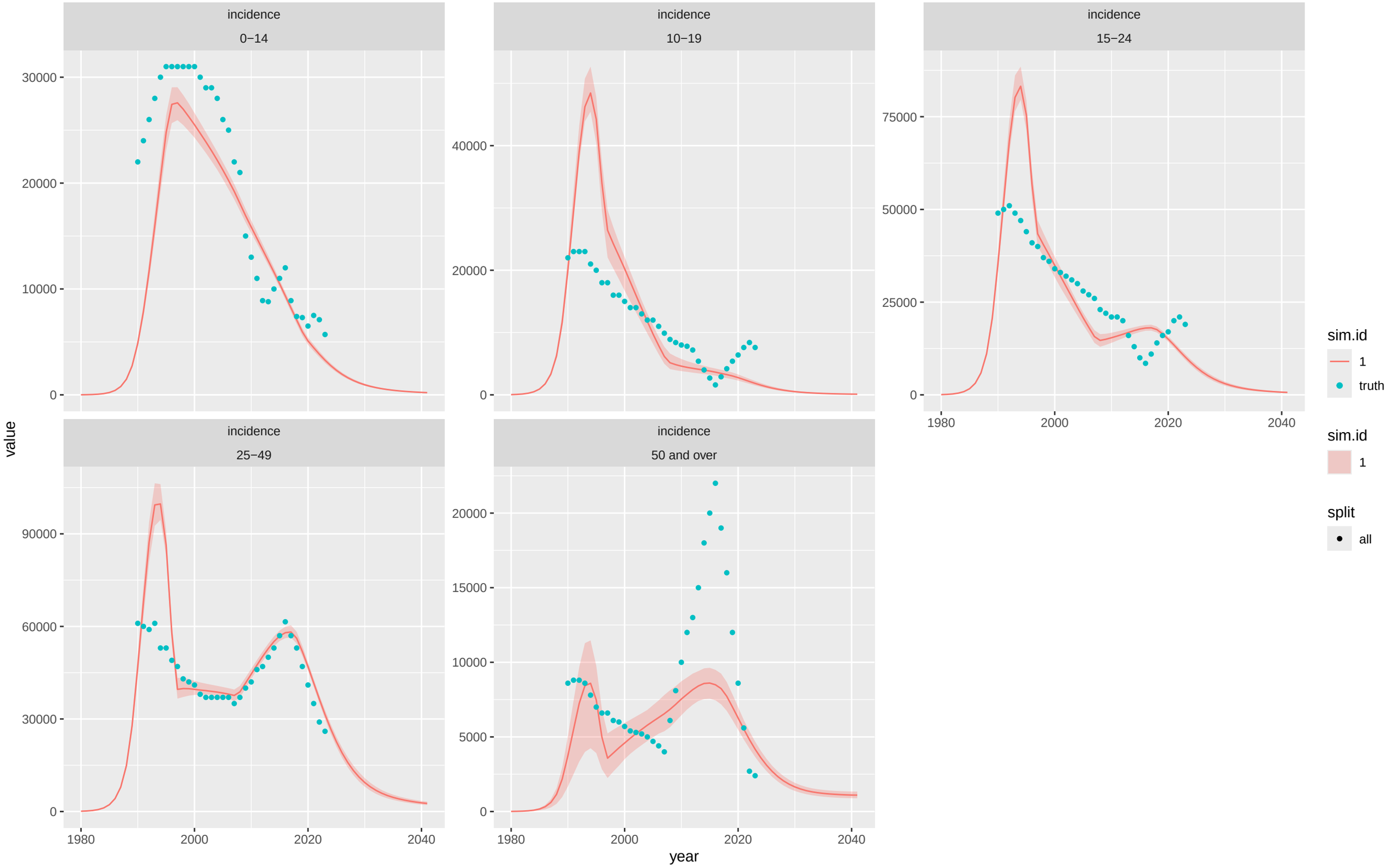
Tanzania step 2: cascade.by.age.sex (one color per chain)



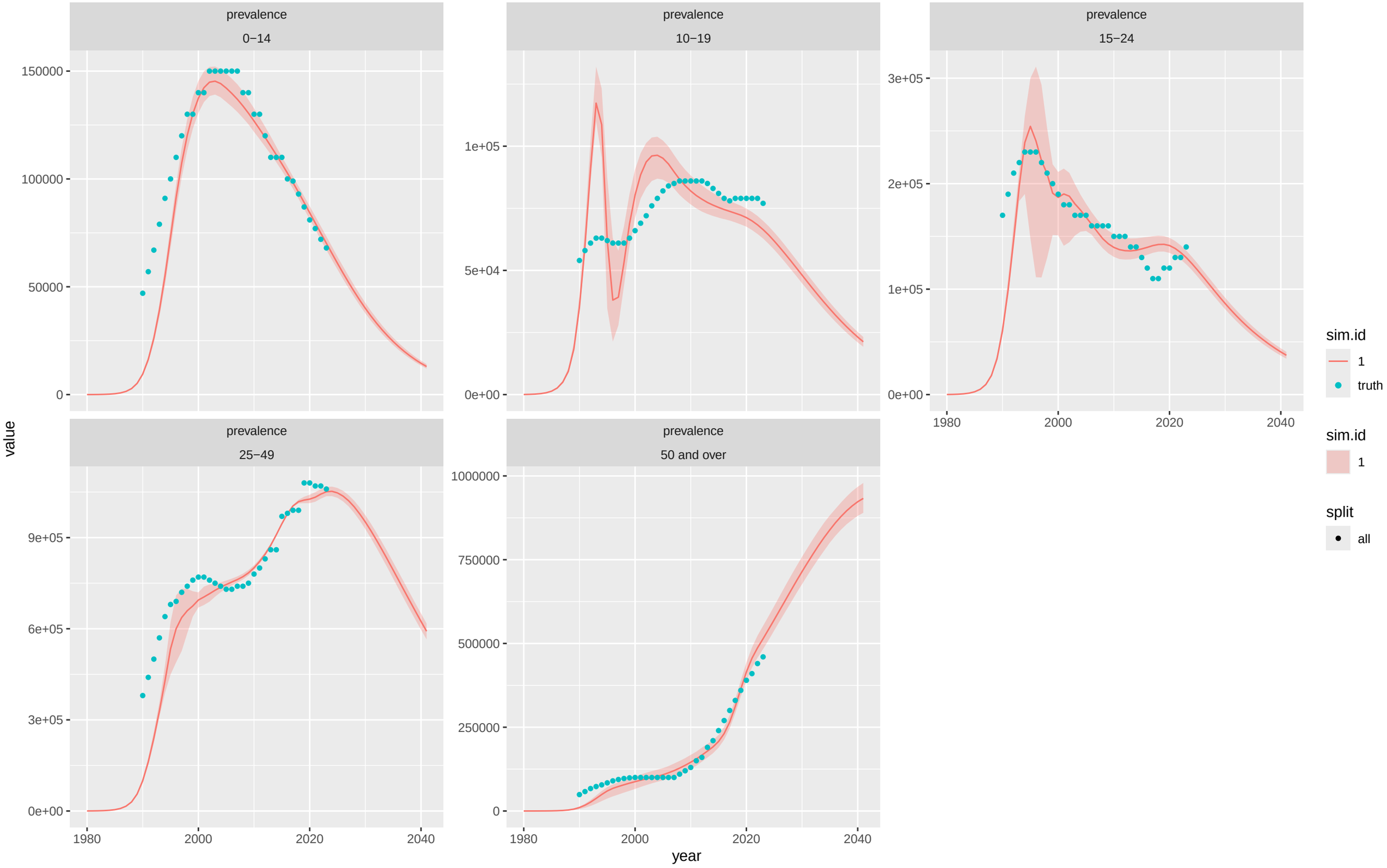
Tanzania: fit (merged Step 2 simset)



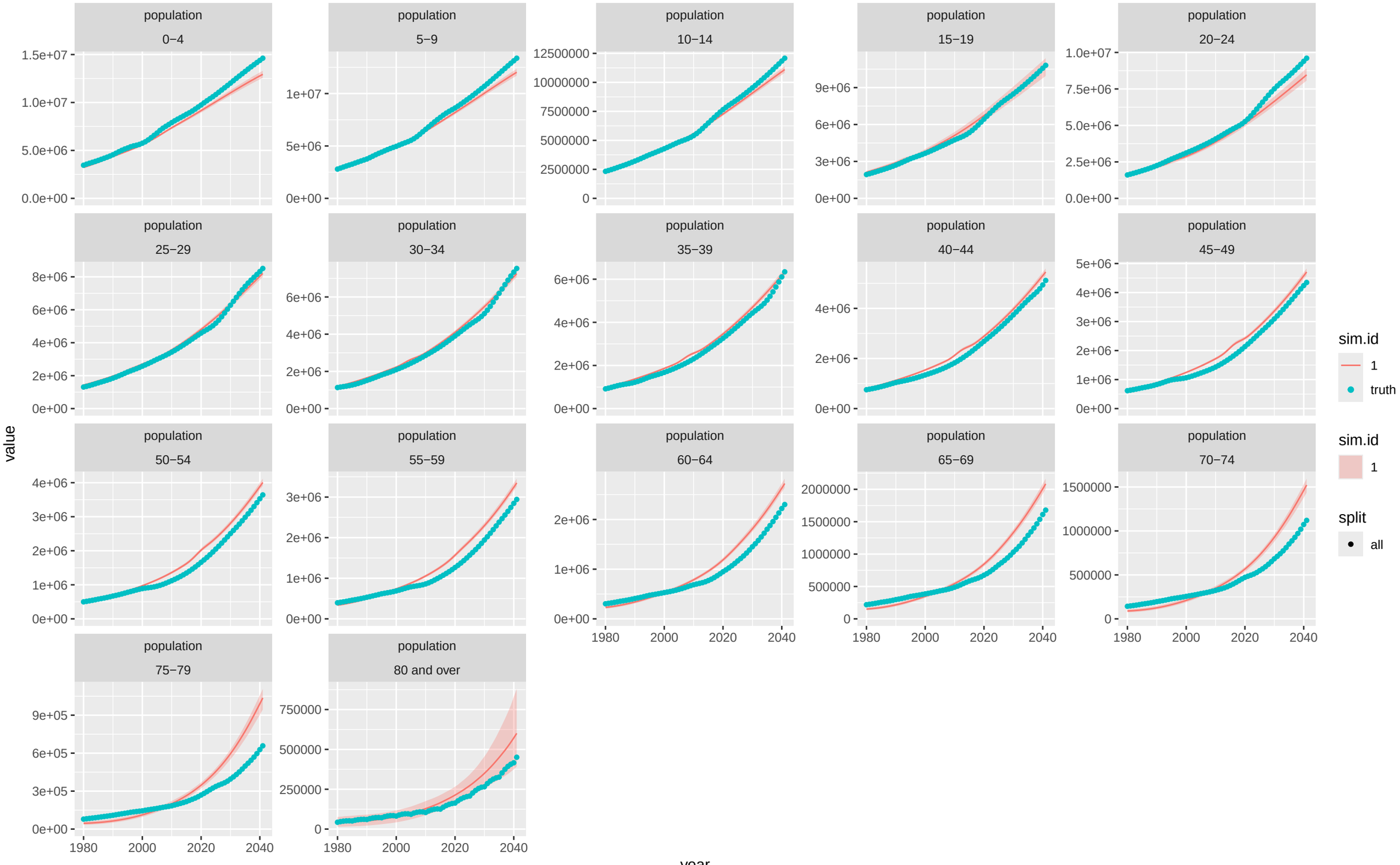
Tanzania: incidence.by.age (merged Step 2 simset)



Tanzania: prevalence.by.age (merged Step 2 simset)



Tanzania: population.by.age (merged Step 2 simset)



Tanzania: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

year

1.00
0.75
0.50
0.25
0.00

1980
2000
2020
2040

sim.id
1
truth

sim.id
1

split
all

Tanzania: cascade.by.age.sex (merged Step 2 simset)

